Supplementary Material to Cognitive Warrant Research

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1. Purpose

This document supplements my research documents, providing background information, intermediate steps, or detailed explanations for those that are interested.

2. Conditional Probability

Fundamental terms of probability

- An **experiment** is any action or process whose outcome is subject to uncertainty.
- The **sample space** is the set of all possible outcomes of that experiment.
- An **event** is any collection (subset) of outcomes contained in the sample space.
- An event is **simple** if it consists of exactly one outcome and **compound** if it consists of more than one outcome.
- A and B are considered **independent** events if the occurrence or nonoccurrence of one event has no bearing on the chance that the other will occur.

We intuitively know that a probability is some measure of an event's likelihood, or the certainty of an outcome, or our degree of belief in a proposition or hypothesis. The probability of an event or outcome is 0, 1, or any number between.

If we have defined an event A, $P(A)$ is the marginal, or unconditional, probability of event A occurring without consideration of any other events. $P(B)$ is thus the marginal probability of event B. If A and B are independent, as defined above, then knowing whether or not B has occurred, or been observed, provides no information about, or has no effect on, the subsequent (non)occurrence of A. If, however, these two events are *not* independent, then the occurrence of B does provide information about, or has an effect on, the subsequent occurrence or observation of A. In this case, the probability of A is conditioned by B and is designated as $P(A \mid B)$.

Definition 0.1 (Conditional probability)**.** Let A and B be two events with $P(B) \neq 0$. The conditional probability of A given B is defined to be

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \tag{1}$$

For example, the (unconditional) probability that a person is wet at any moment might be $P(A) \geq 0$ and the (unconditional) probability that it is raining at that person's location at any moment might be $P(B) \geq 0$. If we know that B has occurred or is occurring, the probability that this person is wet changes. In this situation, the person's state of being drenched is conditional upon B having occurred. We will see below that, on average, conditioning (i.e., knowing the state of one random variable) reduces some uncertainty about another random variable (but leaves that uncertainty unchanged if those random variables are independent). If I know the result of flipping one coin, that does not affect my uncertainty about the result of flipping a different coin; however, if I know that it is raining at a person's location, that reduces my uncertainty (to some degree) about whether he or she is wet.

3. Entropy

A **random variable** is a quantity whose value depends on the outcome of an uncertain process. The name is slightly misleading — it doesn't mean arbitrary or haphazard. It means the value is determined by something not yet observed. A coin flip, a person's response time on a trial, and whether two sentences invoke the same word sense are all random variables: before you observe them, they could take more than one value; after you observe them, they take exactly one. What makes a random variable mathematically tractable is that even though individual outcomes are uncertain, the *distribution* of outcomes — which values are likely and which are unlikely — can be known and reasoned about precisely.

Entropy measures how uncertain that distribution is. More precisely, it quantifies how surprised you should expect to be by the outcome — or equivalently, how much information you gain when you learn it. A fair coin has high entropy: you genuinely don't know what's coming, so the outcome tells you something. A coin weighted to land heads 99% of the time has low entropy: you already know roughly what will happen, so the outcome tells you little. Crucially, entropy is a property of the *distribution*, not of any single observation and not of what the outcomes actually are.

You may be familiar with the term entropy from the second law of thermodynamics and some version of 'over time, everything becomes more disordered.' I wont discuss the use of this term in thermodynamics versus information theory, but perhaps associating disorder with uncertainty will make it easier to understand what is meant by entropy here.

Definition 0.2 (Entropy)**.** Formally, the entropy of the random variable X is

$$H(X) \;=\; -\sum_{x} p(x) \log p(x). \tag{2}$$

For each of the possible values of the random variable, we are multiplying its probability by the logarithm of its probability.[1] The entropy is the negated sum of these products.

Earlier, I said that entropy quantifies how surprised you should expect to be by the outcome of an uncertain process. Surprise is the result of an unexpected event, or, in other words, an event with a low probability. Thus, surprise and likelihood are inversely related: the greater the likelihood of an event, the less surprise when it occurs. A property of logarithms is

$$\log M^k = k \cdot \log M$$

Because we can bring the negative sign into our summation, entropy could be written as

$$\begin{aligned}
H(X) &= \sum_x -p(x)\log p(x) \\
&= \sum_x p(x) \cdot (-1) \cdot \log p(x) \\
&= \sum_x p(x) \cdot \log(p(x))^{-1} \\
&= \sum_x p(x) \cdot \log \frac{1}{p(x)}
\end{aligned}$$

The last term $\frac{1}{p(x)}$ is the inverse of the likelihood of x and thus encodes surprisal in entropy. Taking the logarithm transforms the scale but not the inverse relationship. We can now say that for each possible value of X, we are multiplying the surprise

[1] Note that the base of the logarithm is arbitrary. Base 2 is standard in information theory and the units of entropy are bits for "binary digits."

associated with that value by the value's likelihood. Entropy is the sum of these products.

You may be familiar with a weighted sum. The expected value of a random variable is the sum of its possible values each weighted by their probability. Entropy is thus the expected amount (or weighted average) of surprise over the distribution of the random variable.

An immediate consequence of Definition 0.2 is $H(X) \geq 0$: entropy is always non-negative. If the outcome is certain, $H(X) = 0$ — there is nothing to learn. Otherwise $H(X) > 0$; some uncertainty always remains.

Figure 1 shows entropy for the simplest possible case: a Bernoulli variable that takes only two values (such as same sense / different sense), with probability p for one outcome and $(1 - p)$ for the other. Entropy is zero at the extremes — when the outcome is a near-certainty in either direction — and peaks at $p = \frac{1}{2}$, when both outcomes are equally likely and uncertainty is at its maximum.

4. Conditional Entropy

Conditional entropy measures how much uncertainty remains about one random variable once you already know the value of another. If knowing X tells you a lot about Y, the conditional entropy $H(Y \mid X)$ is low — little uncertainty is left. If knowing X tells you nothing about Y, $H(Y \mid X)$ equals the original entropy of Y — knowledge of X bought you nothing.

The definition computes this by asking: for each possible value of X, how uncertain is Y? It then averages those uncertainties, weighted by how often each value

Binary Entropy Function

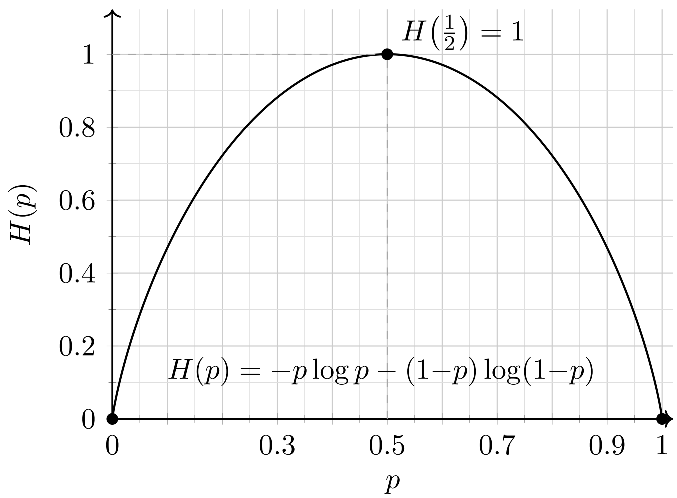

Figure 1: Entropy of a two-outcome variable as a function of the probability p of one outcome. Uncertainty — and therefore entropy — is zero when the result is a near-certainty and is maximized when both outcomes are equally likely.

of X actually occurs:

$$H(Y \mid X) \;=\; \sum_x p(x)\, H(Y \mid X{=}x) = -\sum_x \sum_y p(x,y) \log p(y \mid x). \tag{3}$$

Note. The order matters: knowing X does not generally reduce uncertainty about Y by the same amount that knowing Y reduces uncertainty about X, so $H(Y \mid X) \neq H(X \mid Y)$ in general. However, the *difference* each variable makes is symmetric: $H(X) - H(X \mid Y) = H(Y) - H(Y \mid X)$. Both expressions equal the mutual information between X and Y — the uncertainty each one resolves about the other — introduced in the next section.